Interpretability & Explainability in AI

Mid-Progress Report

AI-Driven Skin Cancer Diagnosis using XAI

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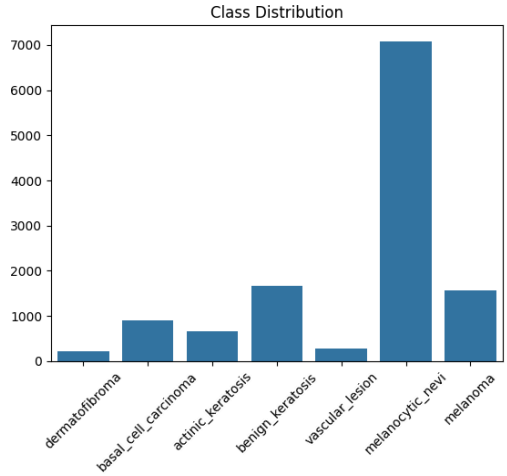
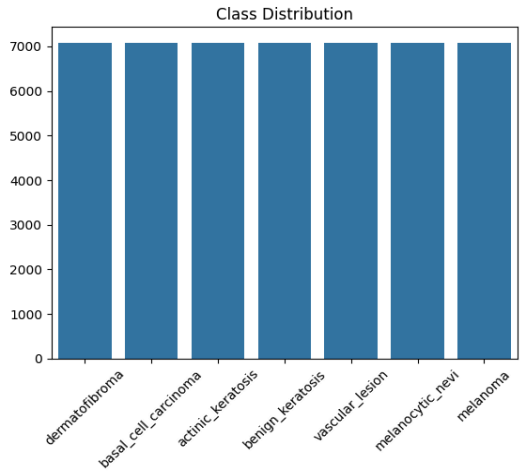
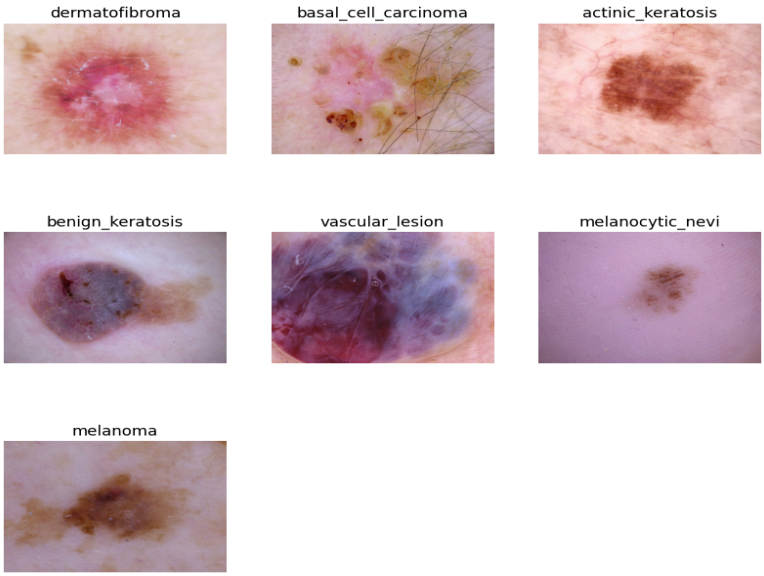
In this project, we aim to build a multiclass skin cancer classification model, specifically a DCNN model. The model is trained on a combination of the HAM10000 dataset and the ISIC skin cancer dataset. The dataset was preprocessed through resizing, normalization, and data augmentation techniques to help improve model performance. In addition, class imbalance was solved using random oversampling.

1. Exploratory Data Analysis (EDA) :

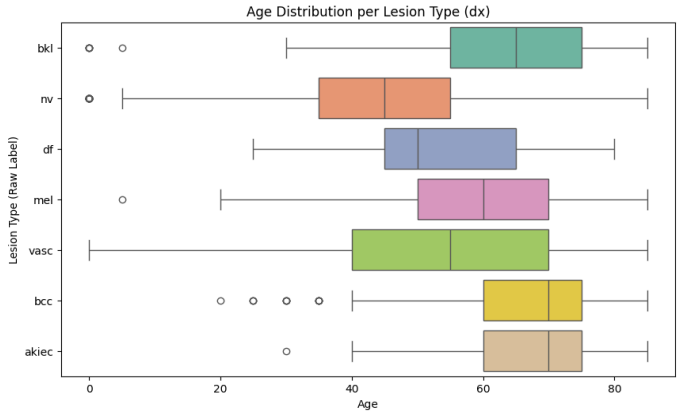




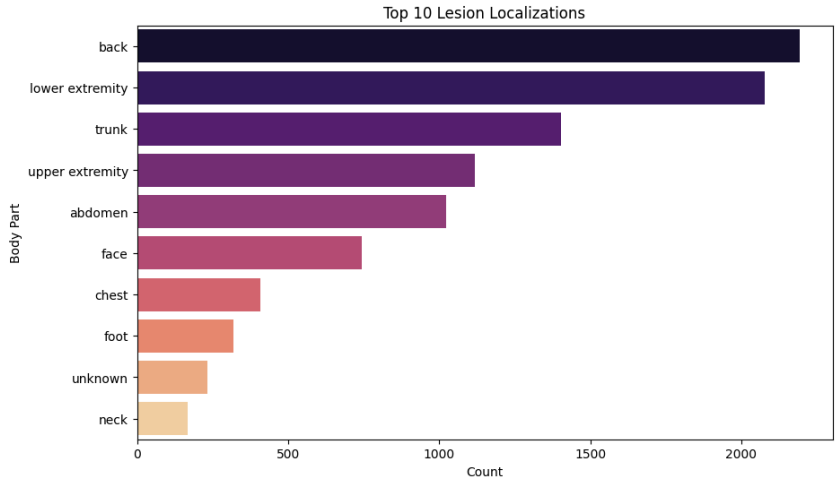
The dataset shows a severe class imbalance, where some classes have more samples than others. This imbalance led the model to be biased toward the majority classes, which leads to poor performance to solve this problem, the oversampling technique was applied during preprocessing.



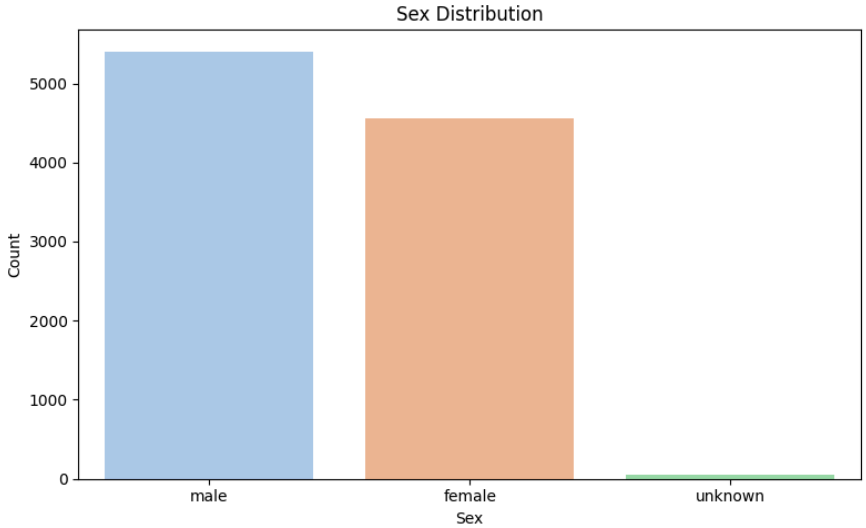
These are sample images for each class; they were visualized to help in showing characteristics of the dataset.



The age distribution per lesion types shows that there are some skin cancer types dominant in specific age groups. which highlights the importance of considering age, as it shows that age is a potential influencing factor in diagnosis.



This graph shows that the distribution of lesion locations indicates that most skin lesions are concentrated in areas such as the back, lower extremities, and trunk.

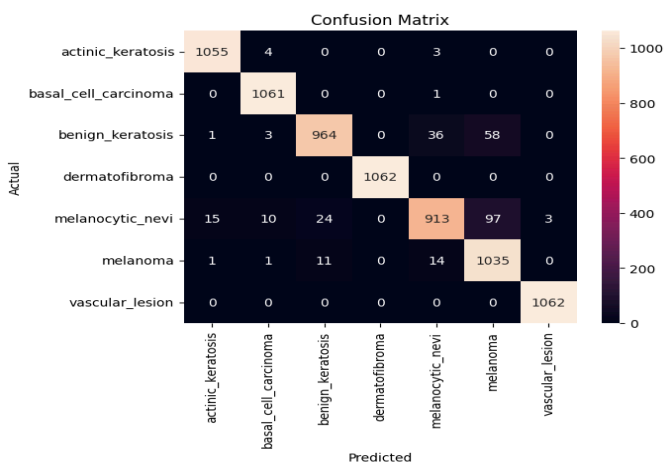


The sex distribution graph shows that the dataset contains a higher number of male samples compared to female samples, while a very small group is unknown. This imbalance indicates that there is a slight gender bias in the dataset.

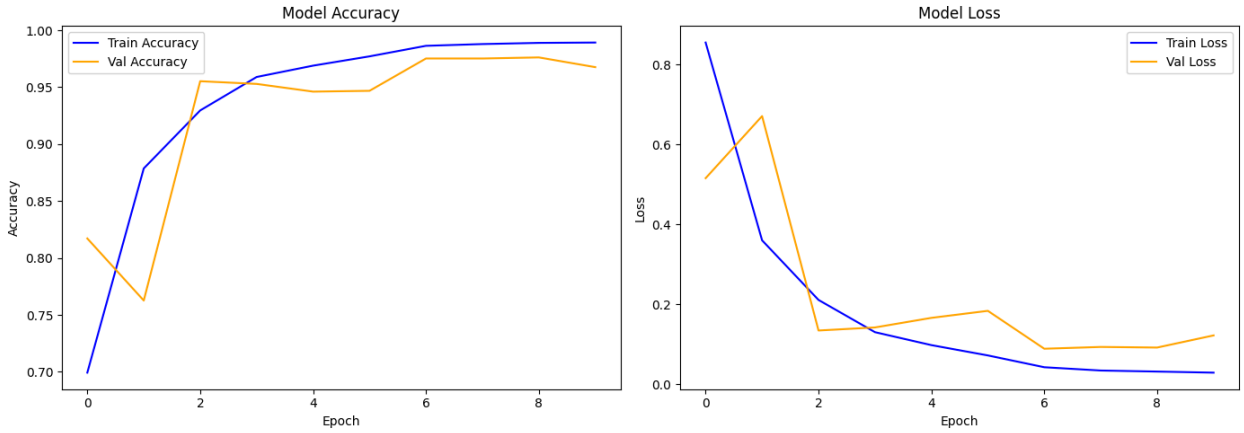
2. Models Results:

```
•••  Accuracy: 0.9620661824051655
                          precision    recall  f1-score   support

       actinic_keratosis       0.98      0.99      0.99      1062
    basal_cell_carcinoma       0.98      1.00      0.99      1062
        benign_keratosis       0.96      0.91      0.94      1062
          dermatofibroma       1.00      1.00      1.00      1062
       melanocytic_nevi        0.94      0.86      0.90      1062
               melanoma       0.87      0.97      0.92      1062
         vascular_lesion       1.00      1.00      1.00      1062

               accuracy                           0.96      7434
              macro avg       0.96      0.96      0.96      7434
           weighted avg       0.96      0.96      0.96      7434
```

This model achieves an overall accuracy 96.2%, which shows strong performance in classifying multiple classes of skin lesion types. There are some high-performing classes, such as **dermatofibroma** and **vascular lesion**, that achieve perfect scores in precision, recall, and F1-score. While **basal cell carcinoma** and **actinic keratosis** also show excellent performance. However, slightly lower performance is observed in **melanocytic nevi** (lower recall) and **melanoma** (lower precision), showing that there is some confusion between visually similar classes. Overall, the model is robust and well-balanced, but can be improved for distinguishing similar lesion types.

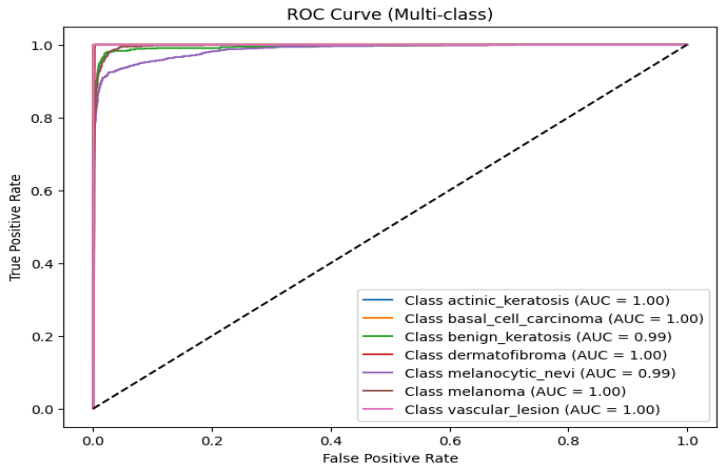


The confusion matrix provides a detailed view of the model's classification performance across all skin lesion classes.

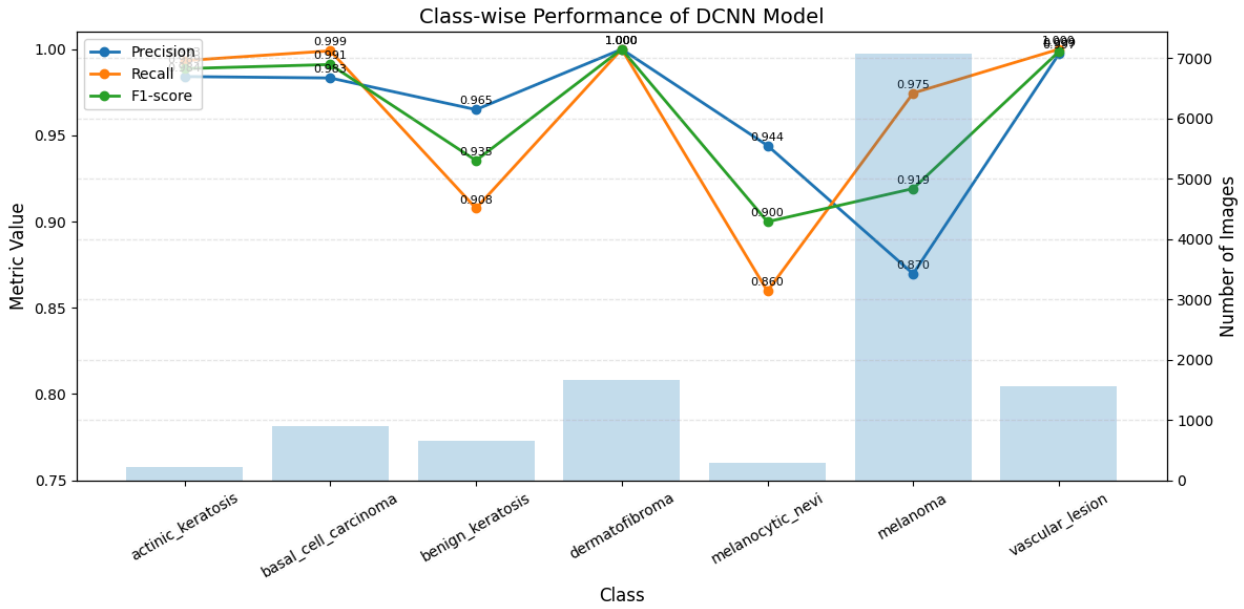


The accuracy curves show that the training accuracy reaches 99% and validation accuracy stabilizes around 96–97%, indicating strong learning and good generalization. The small gap between them suggests minimal overfitting.

Overall, the model demonstrates effective learning with no significant overfitting or underfitting, and shows the suitability of the chosen model architecture and training strategy.

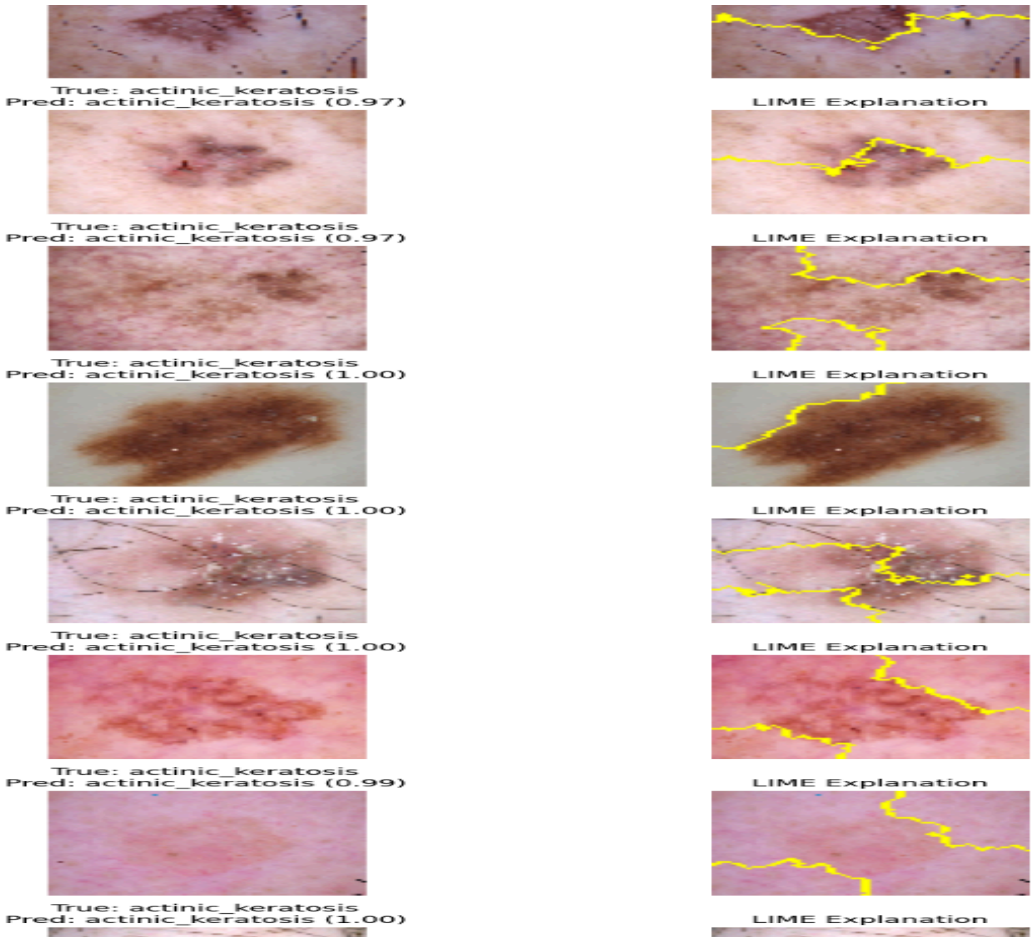


The ROC curves for all classes indicate that the model has a strong ability to distinguish between classes



The class-wise performance shows that most classes achieve high precision, recall, and F1-scores, showing that the model is a strong and balanced classifier.

3.Interpretability techniques and results



The LIME visualizations highlight the most important regions that help in influencing the model's predictions. The highlighted boundaries indicate the areas that contribute most to classifying the lesion